



NP – 410

21  
III Semester B.C.A. Examination, January/February 2025  
(NEP) (F+R)  
**COMPUTER SCIENCE**  
**CA – C13T : Python Programming**

Time : 2½ Hours

Max. Marks : 60

**Instruction :** Answer **any four** questions from **each** Section.

SECTION – A

I. Answer **any four** questions. **Each** question carries **2** marks.

(4×2=8)

- 1) What is Python ? List the execution modes in Python.
- 2) Define function. Name the type of functions in Python.
- 3) What is indexing and negative indexing in a string ?
- 4) Give difference between list and string.
- 5) What is pickling and unpickling ?
- 6) Define abstraction and inheritance.



SECTION – B

II. Answer **any four** questions. **Each** question carries **5** marks.

(4×5=20)

- 7) Explain the features of Python.
- 8) What does it mean that strings are immutable ? Explain five string methods with example.
- 9) Explain len ( ), any ( ), all ( ) and sorted ( ) functions on dictionary.
- 10) Write differences between tuple and dictionary.
- 11) What is data visualization ? Explain role of Matplotlib library in visualization.
- 12) What is frozen set ? Explain difference, difference – update ( ), intersection and intersection – update method on set.

P.T.O.



## SECTION – C

III. Answer **any four** questions. **Each** question carries **8** marks. **(4×8=32)**

- 13) Explain the different types of control flow statements with an example.
  - 14) Describe with example basic operations on list.
  - 15) How to define and call user defined function ? Explain with example.
  - 16) Explain the process of reading and writing a data into file with example.
  - 17) What is a constructor ? What are the types of constructor ? How to define it in Python ?
  - 18) Write a note on :
    - a) Web API
    - b) Plotly.
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